Math 1200 Lecture 20

Nov 12 2026

Topics: 16.7: Scalar and vector surface integrals, flux Name:_____
Collaborators:_____

Group work 1 Let S be the cylinder of radius 2 with open ends at $z = 0$ and $z = 1$ and let $f(x, y, z) = xyz$. Calculate [1] the integral

$$\iint_S f ds$$

[1] recall: $\sin(\theta)\cos(\theta) = \frac{1}{2}\sin(2\theta)$

Group work 2

a) Calculate the flux of the vector field $\vec{F} = \langle x, 2y, z \rangle$ through the portion of the plane $x + y + z = 1$ in the first octant.

b) Let $\vec{F} = \langle x, y, z \rangle$. Let S be the surface $y = \sqrt{x^2 + z^2}$ for $0 \leq y \leq 10$. Calculate

$$\iint_S \vec{F} \cdot d\vec{S}.$$

Hint: Draw the vector field and the surface, you shouldn't have to calculate anything!

Group work 3 Let S be the portion of the graph of $f(x, y) = 1 - x^2 - y^2$ above the xy-plane. Calculate

$$\iint_S \langle x, y, 1 \rangle \cdot d\vec{S}.$$